

Complements of Operational Research

1.3 Probabilistic decision criteria

José Firmino Aguilar Madeira

What if we know the probabilities?

The example from 1.2 (profit in thousands of euros):

	low θ_1	medium θ_2	high θ_3
a_1 : small	40	45	50
a_2 : medium	20	60	80
a_3 : large	-40	50	120
$h(\theta_j)$	0.3	0.5	0.2

The sales department has studied the market and estimates the probabilities in the last row.

Portuguese notation in the tests: VE (= EV), MVE (= MEV), POE (= EOL), VEIP (= EVPI).

What if we know the probabilities?

The example from 1.2 (profit in thousands of euros):

	low θ_1	medium θ_2	high θ_3
a_1 : small	40	45	50
a_2 : medium	20	60	80
a_3 : large	-40	50	120
$h(\theta_j)$	0.3	0.5	0.2

The sales department has studied the market and estimates the probabilities in the last row.

Portuguese notation in the tests: VE (= EV), MVE (= MEV), POE (= EOL), VEIP (= EVPI).

Decision under risk

The probability $h(\theta_j)$ of each state is known, with

$$h(\theta_j) \geq 0, \quad \sum_{j=1}^m h(\theta_j) = 1.$$

What if we know the probabilities?

The example from 1.2 (profit in thousands of euros):

	low θ_1	medium θ_2	high θ_3
a_1 : small	40	45	50
a_2 : medium	20	60	80
a_3 : large	-40	50	120
$h(\theta_j)$	0.3	0.5	0.2

The sales department has studied the market and estimates the probabilities in the last row.

Portuguese notation in the tests: VE (= EV), MVE (= MEV), POE (= EOL), VEIP (= EVPI).

Decision under risk

The probability $h(\theta_j)$ of each state is known, with

$$h(\theta_j) \geq 0, \quad \sum_{j=1}^m h(\theta_j) = 1.$$

Two criteria:

1. maximisation of the **expected value** (MEV);
2. minimisation of the **expected opportunity loss** (EOL).

1. Maximisation of the expected value (MEV)

Steps

1. Assign each state a probability $h(\theta_j)$, with $\sum_j h(\theta_j) = 1$.
2. Expected value of each action:

$$EV_{a_i} = \sum_{j=1}^m h(\theta_j) c_{ij}.$$

3. Choose the action with the largest expected value:

$$MEV = \max_{a_j} EV_{a_j}.$$

1. Maximisation of the expected value (MEV)

Steps

1. Assign each state a probability $h(\theta_j)$, with $\sum_j h(\theta_j) = 1$.
2. Expected value of each action:

$$EV_{a_i} = \sum_{j=1}^m h(\theta_j) c_{ij}.$$

3. Choose the action with the largest expected value:

$$MEV = \max_{a_j} EV_{a_j}.$$

	EV_{a_i}
a_1	$0.3 \times 40 + 0.5 \times 45 + 0.2 \times 50 = 44.5$
a_2	$0.3 \times 20 + 0.5 \times 60 + 0.2 \times 80 = \mathbf{52.0}$
a_3	$0.3 \times (-40) + 0.5 \times 50 + 0.2 \times 120 = 37.0$

$$MEV = \max\{44.5, 52.0, 37.0\} = 52 \Rightarrow$$

a_2 (medium)

1. Maximisation of the expected value (MEV)

Steps

1. Assign each state a probability $h(\theta_j)$, with $\sum_j h(\theta_j) = 1$.
2. Expected value of each action:

$$EV_{a_i} = \sum_{j=1}^m h(\theta_j) c_{ij}.$$

3. Choose the action with the largest expected value:

$$MEV = \max_{a_j} EV_{a_j}.$$

	EV_{a_i}
a_1	$0.3 \times 40 + 0.5 \times 45 + 0.2 \times 50 = 44.5$
a_2	$0.3 \times 20 + 0.5 \times 60 + 0.2 \times 80 = \mathbf{52.0}$
a_3	$0.3 \times (-40) + 0.5 \times 50 + 0.2 \times 120 = 37.0$

$$MEV = \max\{44.5, 52.0, 37.0\} = 52 \Rightarrow$$

a_2 (medium)

What does 52 mean? It is not the profit that will happen (the possible profits of a_2 are 20, 60 or 80): it is the **long-run average**, if the same decision were repeated many times.

2. Minimisation of the expected opportunity loss (EOL)

Steps

1. Compute the regret (opportunity loss) matrix, as in Savage: $r_{ij} = \max_k c_{kj} - c_{ij}$.
2. Expected value of each row:

$$\text{EOL}_{a_i} = \sum_{j=1}^m h(\theta_j) r_{ij}.$$

3. Choose the action with the **smallest** EOL.

2. Minimisation of the expected opportunity loss (EOL)

Steps

1. Compute the regret (opportunity loss) matrix, as in Savage: $r_{ij} = \max_k c_{kj} - c_{ij}$.
2. Expected value of each row:

$$\text{EOL}_{a_i} = \sum_{j=1}^m h(\theta_j) r_{ij}.$$

3. Choose the action with the **smallest** EOL.

	θ_1	θ_2	θ_3	EOL_{a_i}
a_1	0	15	70	$0 + 7.5 + 14 = 21.5$
a_2	20	0	40	$6 + 0 + 8 = \mathbf{14.0}$
a_3	80	10	0	$24 + 5 + 0 = 29.0$
h	0.3	0.5	0.2	

$$\min\{21.5, 14.0, 29.0\} = 14 \Rightarrow \mathbf{a_2 \text{ (medium)}}$$

2. Minimisation of the expected opportunity loss (EOL)

Steps

1. Compute the regret (opportunity loss) matrix, as in Savage: $r_{ij} = \max_k c_{kj} - c_{ij}$.
2. Expected value of each row:

$$\text{EOL}_{a_i} = \sum_{j=1}^m h(\theta_j) r_{ij}.$$

3. Choose the action with the **smallest** EOL.

	θ_1	θ_2	θ_3	EOL_{a_i}
a_1	0	15	70	$0 + 7.5 + 14 = 21.5$
a_2	20	0	40	$6 + 0 + 8 = \mathbf{14.0}$
a_3	80	10	0	$24 + 5 + 0 = 29.0$
h	0.3	0.5	0.2	

$$\min\{21.5, 14.0, 29.0\} = 14 \Rightarrow \mathbf{a_2 \text{ (medium)}}$$

Question: was it a coincidence that both criteria chose the same action?

MEV and EOL always choose the same action

In each row, add the expected value and the expected opportunity loss:

$$EV_{a_i} + EOL_{a_i} = \sum_j h(\theta_j)(c_{ij} + r_{ij}) = \sum_j h(\theta_j) \max_k c_{kj}.$$

The result **does not depend on i** : it is the same for all actions.

	EV	EOL	sum
a_1	44.5	21.5	66
a_2	52.0	14.0	66
a_3	37.0	29.0	66

Consequence

Maximising the EV $\iff$ minimising the EOL. The two criteria **always** give the same decision.

A useful check in the exercises: if the sums EV + EOL are not all equal, there is an arithmetic error.

How much is it worth knowing the future?

If an “oracle” told us what the demand will be, we would always choose the best action in that column: 40 if low, 60 if medium, 120 if high.

Expected value **with perfect information**:

$$0.3 \times 40 + 0.5 \times 60 + 0.2 \times 120 = 66.$$

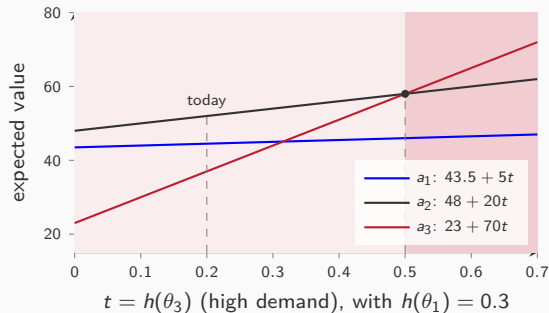
Without the information, the best we can achieve is $MEV = 52$.

Expected value of perfect information

$$EVPI = 66 - 52 = 14 = \min_{a_j} EOL_{a_j}.$$

Application: it is not worth paying more than 14 thousand euros for a market study, however good it is — no information is worth more than perfect information.

Is the decision sensitive to the probabilities?



Fix $h(\theta_1) = 0.3$ and let $t = h(\theta_3)$ vary; then $h(\theta_2) = 0.7 - t$.

Each EV is a straight line in t :

- $t < 0.5$: a_2 (medium);
- $t > 0.5$: a_3 (large).

With the current estimate ($t = 0.2$) there is a margin: the probability of high demand would have to more than double to change the decision.

Before deciding with estimated probabilities, check how wrong they could be.

The expected value does not see risk

With $t = 0.5$ (the boundary), a_2 and a_3 have the same EV (58), but:

- a_2 : possible profits 20, 60, 80 — **never loses**;
- a_3 : possible profits -40, 50, 120 — may give a **loss**.

For whoever makes this decision **only once**, the average is not everything.

In practice

Also look at the **worst case** of each action (like Maximin in 1.2) and at the spread of the results. Using several criteria together is a simple way of doing this.

Formally, risk is handled by replacing the gains with *utilities* — outside the scope of this course.

Example 3 — investment (continued)

	rise	stable	fall
buy shares	1000	0	-1500
buy bonds	350	200	300
buy treasury bills	220	100	0

The market shows probabilities of occurrence of 30% (rise), 50% (stable) and 20% (fall). Which action should be chosen by maximisation of the expected value? And by minimisation of the expected opportunity loss?

Exercise

For the same table, compute the expected value of perfect information. What is the most the investor should pay for an exact forecast of the market?

Check: EV + EOL must give the same value in all three rows.

KEY TAKEAWAYS — 1.3

1. Under **risk**, the probability $h(\theta_j)$ of each state is known (or estimated).
2. **MEV**: $EV_{a_i} = \sum_j h(\theta_j)c_{ij}$; choose the largest. It is a long-run average.
3. **EOL**: the same on the regret matrix; choose the smallest.
4. $EV + EOL$ is the same in every row: **MEV and EOL always give the same decision**.
5. $EVPI = \min EOL$: the most it is worth paying for information.
6. Check the **sensitivity** to the probabilities and do not forget the worst case.

What if the probabilities come from the decision maker's opinion? → 1.4 Subjective probabilities.

Exercise sheet 1 — Exercise 4 (homework)

Ordering umbrellas

A shop has to decide how many umbrellas to order for the season. The result (m.u.) depends on the demand:

Order	strong demand	moderate demand	weak demand
nothing	−40	0	70
40	−10	15	35
80	50	25	−30
160	80	20	−35

Find the decision according to: a) Maximax; b) Maximin; c) Laplace; d) Hurwicz (coefficient of optimism 0.7); e) Savage; f) maximisation of the expected value, with probabilities 30%, 40% and 30%; g) minimisation of the expected opportunity loss, with the same probabilities.

Submission: by email to the lecturer, by the deadline given in class. All exercises: [Exercise sheet 1](#).

References Full list: *Course bibliography*.

- Hillier, F. S., Lieberman, G. J., *Introduction to Operations Research*, 11th ed., McGraw-Hill, 2021 — Decision Analysis (Bayes' decision rule, value of information).
- Taha, H. A., *Operations Research: An Introduction*, 10th ed., Pearson, 2017 — decision under risk.

Resources

- Exercise sheet 1 — Exercises 4 to 9.